

Report of meeting 2

1. Explain how/what parameters were chosen while downloading the MRI images (One University Dataset)

For this research project, we plan to acquire data from the Autism Brain Imaging Data Exchange (ABIDE) repository. To ensure high-quality and consistent data, we specifically focusing on data provided by NYU Langone Medical Center (NYU), which is one of the largest and most reliable institutional contributors to the ABIDE dataset.

When downloading the dataset, the following specific parameters and filters will be chosen to align with the proposed deep learning methodology:

- **Imaging Modality (T1-weighted Structural MRI):** We exclusively plan to select high-resolution structural MRI (sMRI) scans, specifically T1-weighted images.
- **Preprocessing Pipeline (Derivatives):** We plan to select data that has undergone standardized preliminary preprocessing steps. This includes brain extraction (skull-stripping using FSL BET) and spatial normalization, ensuring that non-brain tissues (skull, scalp, dura) are removed prior to deep learning feature extraction.
- **Phenotypic Availability:** A critical selection parameter is the inclusion of complete clinical phenotypic data. We specifically want to filter for the subjects that have complete records for Age at Scan, Sex, Full-Scale IQ (FIQ), Verbal IQ (VIQ), and Performance IQ (PIQ). This allowed us to construct a multi-modal neural network that fuses structural MRI imaging features with clinical tabular data.
- **Diagnostic Cohort:** The dataset will be balanced to include an equal distribution of subjects diagnosed with Autism Spectrum Disorder (ASD) and Neurotypical Control (TC) subjects.

2. What kind of file is required for the MRI analysis?

Unlike standard 2D image formats (like .jpg or .png), a NIfTI file is required for medical analysis because it contains two critical components:

1. **3D Volumetric Data:** It stores the actual 3-dimensional array of voxels (3D pixels) representing the brain's anatomy across the axial, sagittal, and coronal planes.

2. The Header & Affine Matrix: NIfTI files contain metadata that maps the 3D voxel coordinates to real-world spatial coordinates (in millimeters). It stores critical medical parameters such as voxel dimensions, scanner orientation, slice thickness, and magnetic field properties, ensuring the brain is anatomically scaled correctly regardless of the MRI scanner used.

3. Parameters in MRI: Write a small Python code to display various parameters of MRI

To interact with and extract parameters from .nii files in Python, we plan to use the nibabel neuroimaging library. The following script loads a patient's MRI scan and extracts its core physical and spatial parameters directly from the NIfTI header.

Code:

```
import nibabel as nib
import numpy as np
def display_mri_parameters(nifti_path):
    print(f"Loading MRI scan from: {nifti_path}\n")
    try:
        # Load the NIfTI file
        img = nib.load(nifti_path)

        # 1. 3D Image Shape (Number of voxels in X, Y, Z planes)
        shape = img.shape
        print(f"1. 3D Volume Shape (X, Y, Z): {shape} voxels")

        # 2. Voxel Size / Zoom (Physical size of each voxel in millimeters)
        header = img.header
        zooms = header.get_zooms()
        print(f"2. Voxel Dimensions (mm): {zooms}")

        # 3. Data Type (e.g., float32, int16 - determines intensity precision)
        data_type = header.get_data_dtype()
        print(f"3. Pixel Intensity Data Type: {data_type}")
```

```
# 4. Affine Matrix (Maps voxel coordinates to real-world physical space)
affine = img.affine
print(f"\n4. Spatial Affine Transformation Matrix:\n{np.array_str(affine,
precision=2, suppress_small=True)}")
```

```
# 5. Extract actual pixel intensity statistics from the volume array
volume_data = img.get_fdata()
print(f"\n5. Intensity Statistics:")
print(f" - Max Brightness: {np.max(volume_data):.2f}")
print(f" - Min Brightness: {np.min(volume_data):.2f}")
print(f" - Mean Brightness: {np.mean(volume_data):.2f}")
except Exception as e:
    print(f"Error loading MRI file: {e}")
```

Example Usage:

```
# display_mri_parameters("patient_005001.nii.gz")
```

Expected Output Example from the Code:

```
Loading MRI scan from: patient_005001.nii.gz
1. 3D Volume Shape (X, Y, Z): (256, 256, 170) voxels
2. Voxel Dimensions (mm): (1.0, 1.0, 1.2)
3. Pixel Intensity Data Type: int16
4. Spatial Affine Transformation Matrix:
[[-1.  0.  0. 128.]
 [ 0.  1.  0. -128.]
 [ 0.  0.  1.2 -85.]
 [ 0.  0.  0.  1.]]
5. Intensity Statistics:
  - Max Brightness: 2450.00
  - Min Brightness: 0.00
  - Mean Brightness: 412.35
```